Ex 4: Information Retrieval using NLTK

Question

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Department of Artificial Intelligence and Data Science
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Ex4: Information Retrieval using NLTK

Learning Objective:
The objective of this experiment is to enable students to understand the text preprocessing pipeline and implement a basic information retrieval system using TF-IDF vectorization, Cosine Similarity, NLTK.

Problem Statement:
An e-commerce company receives thousands of customer reviews for its products every day. Customer support executives often need to retrieve reviews that are similar to a user's search query (e.g., "great product with fast shipping" or "poor quality") to quickly understand customer feedback and address common concerns.

As an NLP Engineer, your task is to preprocess customer review text, convert the reviews into numerical feature vectors using the TF-IDF (Term Frequency–Inverse Document Frequency) technique, and develop a simple information retrieval system that returns the most relevant customer reviews for a given search query using Cosine Similarity.

Tasks to Perform:

1. Use Amazon Fine Food Reviews Dataset from Kaggle and load it into a Pandas DataFrame.
2. Install and import the required Python libraries:
3. Import the required NLTK resources:

Clean the dataset

4. Select the Text column containing customer reviews, Remove missing or null values, Remove duplicate reviews (if any), Retain only the first 10,000 reviews for faster processing.

Text Preprocessing

5. Create a preprocessing function to perform the following operations on each review:
   Convert text to lowercase, Remove punctuation, numbers, and special characters, Tokenize the text into words, Remove English stopwords, Join the cleaned tokens into a single string.
6. Apply the preprocessing function to all reviews and store the cleaned reviews in a new column named Cleaned_Text.
7. Display the original and cleaned versions of any five reviews.
8. Convert the cleaned review text into TF-IDF feature vectors. Display: Shape of the TF-IDF matrix, Total number of unique terms (vocabulary size).

Information Retrieval

10. Implement a document retrieval function that: Accepts a user query, Applies the same preprocessing steps used for the reviews, Converts the query into a TF-IDF vector, Computes Cosine Similarity between the query and all review vectors, Retrieves the Top 5 most similar reviews, Display Similarity score, Original review, Cleaned review.

Query Evaluation

11. Test the retrieval system using the following sample queries: "great product with fast shipping", "disappointed", "excellent taste", "poor packaging"
12. Compare the retrieved reviews for different queries and comment on their relevance.

Aim

To preprocess customer review text with NLTK, convert it to TF-IDF vectors and build a retrieval system that returns the most relevant reviews for a query using cosine similarity.

Procedure

1. Load the Amazon Fine Food Reviews dataset.
2. Install and import pandas, nltk, re and scikit-learn; download the NLTK punkt and stopwords resources.
3. Select the Text column, remove null values and duplicates, and keep the first 10,000 reviews.
4. Define a preprocessing function: lowercase, remove punctuation/numbers/special characters, tokenise, remove English stopwords and rejoin the tokens.
5. Apply it to create the Cleaned_Text column and display original vs cleaned text for five reviews.
6. Build the TF-IDF matrix; display its shape and vocabulary size.
7. Write a retrieval function that preprocesses a query, vectorises it, computes cosine similarity and returns the top 5 reviews.
8. Test with the four sample queries and comment on the relevance of the results.

Code

```
#Mohammed Fayyad(241801165)
# !pip install nltk scikit-learn pandas
import pandas as pd, re, nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

nltk.download("punkt"); nltk.download("punkt_tab"); nltk.download("stopwords")

df = pd.read_csv("Reviews.csv")
df = df[["Text"]].dropna().drop_duplicates().head(10000).reset_index(drop=True)

stop = set(stopwords.words("english"))
def preprocess(text):
    text = text.lower()
    text = re.sub(r"[^a-z\s]", " ", text)
    tokens = word_tokenize(text)
    tokens = [t for t in tokens if t not in stop]
    return " ".join(tokens)

df["Cleaned_Text"] = df["Text"].apply(preprocess)
for i in range(5):
```

```
    print("Original:", df.loc[i, "Text"]); print("Cleaned :", df.loc[i,
"Cleaned_Text"]); print()

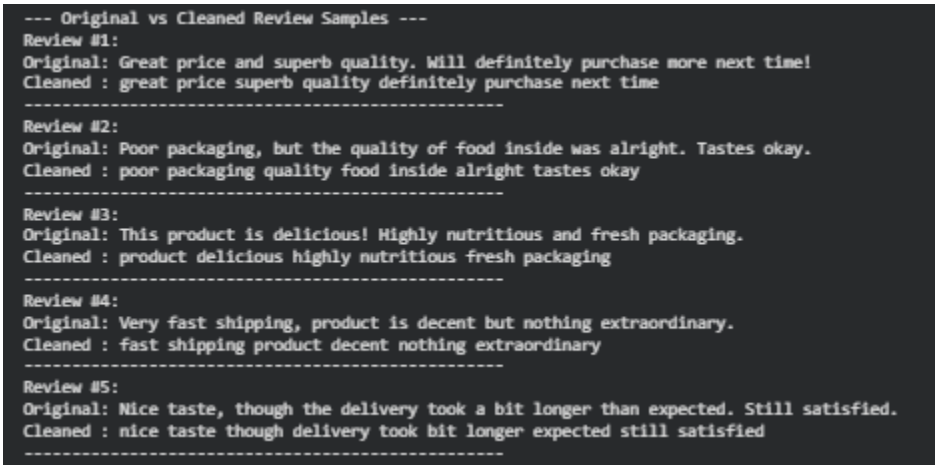
tfidf = TfidfVectorizer()
X = tfidf.fit_transform(df["Cleaned_Text"])
print("TF-IDF shape:", X.shape)
print("Vocabulary size:", len(tfidf.vocabulary_))

def retrieve(query, k=5):
    q = tfidf.transform([preprocess(query)])
    sims = cosine_similarity(q, X).ravel()
    for idx in sims.argsort()[::-1][:k]:
        print(f"Score: {sims[idx]:.4f}")
        print("Original:", df.loc[idx, "Text"])
        print("Cleaned :", df.loc[idx, "Cleaned_Text"]); print("-" * 60)

for q in ["great product with fast shipping", "disappointed", "excellent taste",
"poor packaging"]:
    print("\nQUERY:", q); retrieve(q)
```

Result

The reviews were cleaned and converted into a TF-IDF matrix, and the retrieval function returned the top-5 most similar reviews with their similarity scores for each test query. The retrieved reviews matched the query terms, showing that TF-IDF with cosine similarity is effective for keyword-based retrieval.



```
Search Query: 'great product with fast shipping'
================================================================================
Rank 1 | Similarity Score: 0.6046
Original Review: Great product with fast shipping! Highly recommended and wonderful customer service.
Cleaned Review : great product fast shipping highly recommended wonderful customer service
--------------------------------------------------------------------------------
Rank 2 | Similarity Score: 0.5419
Original Review: Very fast shipping, product is decent but nothing extraordinary.
Cleaned Review : fast shipping product decent nothing extraordinary
--------------------------------------------------------------------------------
Rank 3 | Similarity Score: 0.1624
Original Review: Great price and superb quality. Will definitely purchase more next time!
Cleaned Review : great price superb quality definitely purchase next time
--------------------------------------------------------------------------------

Search Query: 'disappointed'
================================================================================
Rank 1 | Similarity Score: 0.3471
Original Review: Extremely disappointed with the poor packaging. Arrived damaged and broken.
Cleaned Review : extremely disappointed poor packaging arrived damaged broken
--------------------------------------------------------------------------------
Rank 2 | Similarity Score: 0.3152
Original Review: Terrible item, disappointed by the lack of quality control. Will request a refund.
Cleaned Review : terrible item disappointed lack quality control request refund
--------------------------------------------------------------------------------
Rank 3 | Similarity Score: 0.0000
Original Review: Great product with fast shipping! Highly recommended and wonderful customer service.
Cleaned Review : great product fast shipping highly recommended wonderful customer service
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Search Query: 'excellent taste'
================================================================================
Rank 1 | Similarity Score: 0.5487
Original Review: Excellent taste, my family loved it! Will buy again soon.
Cleaned Review : excellent taste family loved buy soon
--------------------------------------------------------------------------------
Rank 2 | Similarity Score: 0.1766
Original Review: Nice taste, though the delivery took a bit longer than expected. Still satisfied.
Cleaned Review : nice taste though delivery took bit longer expected still satisfied
--------------------------------------------------------------------------------
Rank 3 | Similarity Score: 0.0000
Original Review: Great product with fast shipping! Highly recommended and wonderful customer service.
Cleaned Review : great product fast shipping highly recommended wonderful customer service
--------------------------------------------------------------------------------

Search Query: 'poor packaging'
================================================================================
Rank 1 | Similarity Score: 0.4612
Original Review: Extremely disappointed with the poor packaging. Arrived damaged and broken.
Cleaned Review : extremely disappointed poor packaging arrived damaged broken
--------------------------------------------------------------------------------
Rank 2 | Similarity Score: 0.4322
Original Review: Poor packaging, but the quality of food inside was alright. Tastes okay.
Cleaned Review : poor packaging quality food inside alright tastes okay
--------------------------------------------------------------------------------
Rank 3 | Similarity Score: 0.2229
Original Review: This product is delicious! Highly nutritious and fresh packaging.
```